

$$\begin{aligned}
 & c) xy + x'y + xy' \\
 & = xy + (xy + x'y) \text{ (Postulate 3(a))} \\
 & = (xy + xy) + x'y \text{ (Theorem 4(a))} \\
 & = xy(z + z') + x'y \text{ (Postulate 4(a))} \\
 & = xy(z + z') + x'y \text{ (Postulate 3(a))} \\
 & = xy \cdot 1 + x'y \text{ (Postulate 5(a))} \\
 & = xy + x'y \text{ (Postulate 2(b))} \\
 & = yx + yx' \text{ (Postulate 3(b))} \\
 & = y(x + x') \text{ (Postulate 4(a))} \\
 & = y \cdot 1 \text{ (Postulate 5(a))} \\
 & = y \text{ (Postulate 2(b))}
 \end{aligned}$$

$$\begin{aligned}
 & d) zx + zx'y \\
 & = zx + z \cdot (x'y) \text{ (Theorem 4(b))} \\
 & = z(x + x'y) \text{ (Postulate 4(a))} \\
 & = z((x + x') \cdot (xy)) \text{ (Postulate 4(b))} \\
 & = z(1 \cdot (xy)) \text{ (Postulate 5(a))} \\
 & = z(xy) \text{ (Postulate 3(b))} \\
 & = z(xy) \text{ (Postulate 2(b))} \\
 & = zx + zy \text{ (Postulate 4(a))}
 \end{aligned}$$

$$\begin{aligned}
 & e) (A+B)' \cdot (A'+B')' = ((A+B) + (A'+B'))' \text{ (Theorem 5(a))} \\
 & = ((A+B) + (B'+A'))' = (A + (B + (B'+A')))' \text{ (Theorem 4(a))} \\
 & \quad \text{(Postulate 3(a))} = (A + (B+B') + A'))' \text{ (Theorem 4(a))} \\
 & = (A + (HA'))' \text{ (Postulate 5(a))} = (A + (A'+1))' \text{ (Postulate 3(a))} \\
 & = (A+1)' \text{ (Theorem 2(a))} = 1' \text{ (Theorem 2(a))} \\
 & = 0 \text{ (by defn.)}
 \end{aligned}$$

$$\begin{aligned}
 & f) y(wz' + wz) + xy \\
 & = y \cdot (w(z' + z)) + xy \text{ (Postulate 4(a))} \\
 & = y \cdot w \cdot 1 + xy \text{ (Postulate 5(a))} \\
 & = yw + xy \text{ (Postulate 2(b))} \\
 & = wy + xy \text{ (Postulate 3(b))}
 \end{aligned}$$